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A Parametric Analysis of Orbital Debris Collision Probability and Maneuver Rate for Space Vehicles

Mission Operations Directorate
Flight Design and Dynamics Division

Engineering Directorate
Navigation, Control, and Aeronautics Division

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By:

James Lee Foster, Jr.
Barrios Technology, Inc.
Mission Operations Directorate

Herbert S. Estes
Engineering Directorate

Approved by:

Flight Design and
Dynamics Division

Michael E. Evans
Section Head, Advanced Program
Integration

Emery E. Smith
Chief, Flight Analysis Branch

Jon C. Harpold
Chief, Flight Design and
Dynamics Division

Navigation, Control, and
Aeronautics Division

William L. Jackson
Section Head, Guidance and
Proximity Operations

Gregory C. Hite
Chief, Navigation and Guidance
Systems Branch

Kenneth J. Cox
Chief, Navigation, Control, and
Aeronautics Division

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1. Introduction

This paper documents a parametric analysis of collision probability between low Earth orbit space vehicles and orbital debris. Its purpose is to understand the probability of debris impact and assess how that collision probability can be managed through debris avoidance maneuvers by a space vehicle over a period of time.

No matter how high the frequency of debris avoidance maneuvers, probability of debris collision can never be completely eliminated. The probability of collision over time, predicted state vector accuracies for debris and for a space vehicle, and maneuver rate for any space vehicle in low Earth orbit are fundamentally interrelated.

This work discusses the rationale for the selection of the values of analysis parameters, the methods of determining hit probability for a close passage (conjunction) between a space vehicle and a debris object, the method of determining maneuver rate and remaining collision probability, and the significance of the maneuver rate versus collision probability curves. Assuming the Kessler orbital debris environment model (Kessler [1]) (refer to appendix A) and specific space vehicle dimensions, estimated residual collision probability and maneuver rates are calculated.

2. Discussion of Parameters

This section gives the specific values of the analysis parameters chosen and the justification for this selection of values. Specifically, state vector uncertainty for a space vehicle at time of conjunction, state vector uncertainty for a debris object at time of conjunction, the collision probability used for a maneuver decision, the maneuver frequency and the debris flux model are examined.

2.1 State Vector Uncertainty for a Space Vehicle

Earth orbiting space vehicles are currently tracked through the use of the Tracking and Data Relay Satellite System (TDRSS) transponder, radar skin tracking (C-Band), and the Global Positioning System (GPS). The tracking accuracies for TDRSS transponder and C-Band are comparable.

The Space Shuttle is currently tracked by TDRSS transponder and C-band (5 cm wavelength) ground-based radar stations. The selection of space vehicle position uncertainties for this work was based on a memo (Peterson [2]) on typical Shuttle tracking accuracies. Chosen from the memo were the uncertainties associated with an 80-foot semimajor axis error at epoch, propagated for 2, 4, and 8 orbits. The 80-foot semimajor axis uncertainty corresponds to a low but not zero activity space vehicle. Based on these uncertainties, the normally distributed 1σ UVW space vehicle position uncertainties are arbitrarily chosen as:

.12 km x 0.6 km x .12 km	≈ (400 ft x 2000 ft x 400 ft)
.12 km x 1.2 km x .12 km	≈ (400 ft x 4000 ft x 400 ft)
.12 km x 2.4 km x .12 km	≈ (400 ft x 6000 ft x 400 ft)

In addition, the perfect accuracy case for the space vehicle is included in the analysis to examine the high accuracy limit situation. These values are assumed to span the range of state vector uncertainties for a space vehicle.

2.2 State Vector Uncertainty for Tracked Orbital Debris

Orbiting objects with large radar cross-sections ($\geq$ 100 cm^2) are tracked by the United States Space Command (USSPACECOM). The USSPACECOM network of radar sensors operates on L-p band wavelengths of about 70 cm. Because of diffraction effects, this wavelength limits the smallest-sized object which USSPACECOM can reliably track to about 10 cm.

USSPACECOM maintains a database of orbiting objects and routinely updates their state vectors. For routine tracking, USSPACECOM uses the efficient long-term propagators, the SDP4 and SGP4 processors, for deep space and low Earth orbit respectively. If additional position accuracy is necessary, USSPACECOM can perform high-frequency tracking and process data with a "special perturbation" (SP) processor, which integrates the orbit using high-fidelity geopotential and atmosphere models and accounts for lunar-solar perturbations. The SP processor, when used with sufficient tracking data, produces state vectors comparable in accuracy with TDRSS and C-band tracking. The state vector uncertainties chosen for the debris objects are based upon that for the Space Shuttle presented above, and a U- to V- to W-uncertainty ratio inferred from a USSPACECOM communication. The orbital debris UVW normally distributed 1σ-position uncertainties are chosen for this analysis as:

.12 km x 0.6 km x .12 km	≈ (400 ft x 2000 ft x 400 ft)
.24 km x 1.2 km x .24 km	≈ (800 ft x 4000 ft x 800 ft)
.48 km x 2.4 km x .48 km	≈ (1600 ft x 8000 ft x 1600 ft)
.96 km x 4.8 km x .96 km	≈ (3200 ft x 16000 ft x 3200 ft)

These values are assumed to span the range of tracking accuracies for USSPACECOM.

2.3 Collision Probability for a Maneuver Decision

United States space program projects have historically accepted levels of risk for contingency situations of 10^{-4} to 10^{-5} (Sawin et al [3].) For this paper, the acceptable level of risk for penetration of a sphere about the space vehicle by an orbital debris object is allowed to range from 10^{-3} to 10^{-5}. It is assumed that a maneuver is executed whenever the probability of collision equals or exceeds the acceptable level of risk.

2.4 Maneuver Frequency

There is a limit on the frequency of maneuvers for every space vehicle which depends on the mission objectives and the particular vehicle characteristics, such as propulsion system hardware constraints. For example, a commercial communications satellite has operational constraints which limit its maneuver rate. The maneuver rate is determined by the need to protect the satellite, while conserving non-renewable propellant. Likewise, the Space Station is designed to function as a microgravity laboratory, and currently has a requirement for six 30-day microgravity (maneuver-free) periods per year (see appendix F). Every operational space vehicle is subject to some constraint on maneuver frequency. Thus, maneuver rate must be considered when performing a debris avoidance analysis.

2.5 Debris Flux Model

Kessler et al have statistically modeled the orbital debris environment (Kessler [1]). This model predicts the number of debris objects passing through a unit area (flux) as a function of time. The Kessler model is primarily based upon the debris population tracked by USSPACECOM, ground telescope data, and measurements of debris impacts on satellites. This model produces a debris flux as a function of 1) the 10.7 cm solar radiation level averaged over a 13-month period, 2) the altitude of a space vehicle, 3) the orbital inclination of a space vehicle, and 4) the angle of incidence of debris to an orbiting space vehicle.

The current 1992 USSPACECOM database includes about 5% of the predicted larger-than-1-cm population, according to the Kessler Orbital Debris Flux Model (Kessler[1]). While there are far more larger-than-1-cm objects untracked than tracked, these objects comprise a small fraction of the total orbiting mass. Approximately 3,000,000 kg of mass are tracked by USSPACECOM while about 1000 kg of mass are untracked (Kessler[4]). The average weight of the tracked objects is about 1000 pounds; the average weight of untracked debris objects over 1 cm in size is on the order of 1/4 oz.

Figure 1 shows the cumulative probability of debris impact for a 2000 m^2 cross-section space vehicle (about space station size) over a 30-year period for 1997 to 2027. This probability is computed from the Kessler model due to the flux of all objects, tracked and untracked, larger than 10 cm. The model predicts that about 37% of the objects larger than 10 cm are actually tracked. The Kessler Model is described in more detail in appendix A. The calculation assumes no debris avoidance maneuvers, a constant 215 nautical mile altitude, and a constant 10.7 cm solar flux of 70 x 10^{-22} Watts m^{-2} Hz^{-1}. This solar flux corresponds to low solar activity and will result in a higher cumulative collision probability than might actually be expected.

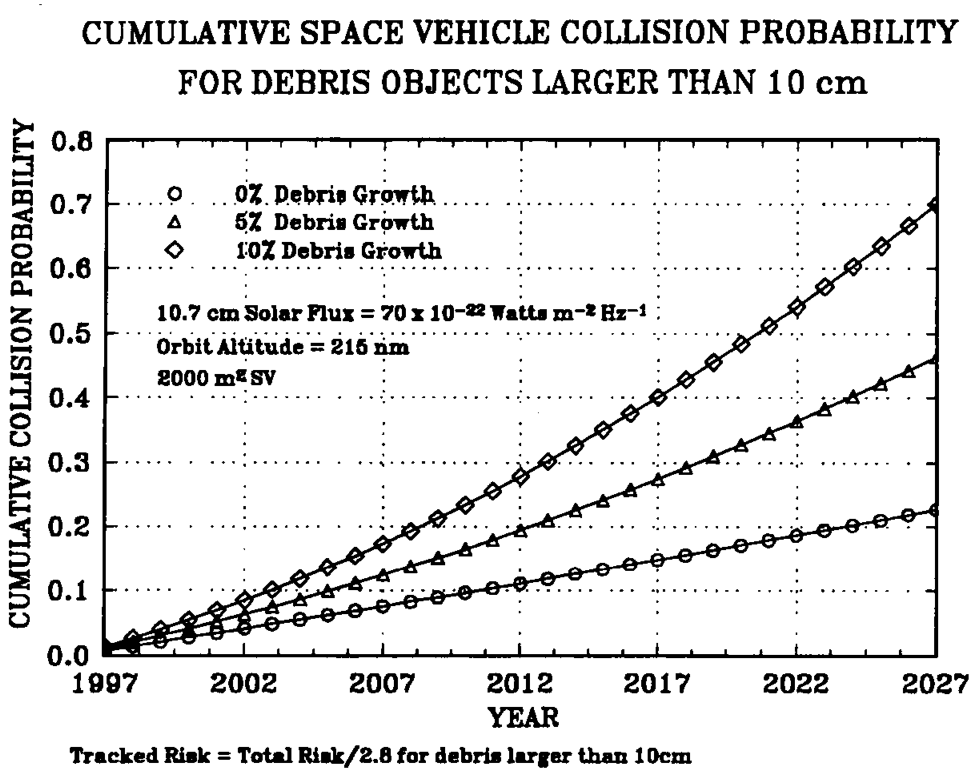

Figure 1. Cumulative risk to a 2000 m^2 space vehicle over a 30-year period due to the debris environment model flux of all objects, larger than 10 cm.

3. Analysis

In this section an overview of the assumptions, tools, and methodology used for this analysis is presented.

3.1 Assumptions and Rationale

This analysis makes the following assumptions:

1. The shapes (U-V-W ratios) of the debris state vector uncertainty ellipsoids presented in section 2.2 are representative of those for the objects in the actual debris population.
2. All debris objects move parallel to the Earth at the same velocity as in the Kessler Model (appendix A).
3. The meteoroid flux is negligible in comparison to the debris flux.
4. All errors are assumed to be normally distributed.

The shapes of the debris state vector uncertainty ellipsoids are selected on the basis of very limited information and must be considered as an assumption. About 98% of the debris population is adequately described as moving parallel to the earth (Anz-Meador [5]) and the analysis slightly over-estimates the collision probability over time.

3.2 Tools and Data

The facilities used for this study include the Johnson Space Center (JSC) Engineering Directorate Guidance, Navigation, and Control (GN&C) emulation laboratory with an array of fifteen 80386 personal computers. Preliminary work was done with the Mission Operations Directorate HP9000 computers using a program developed by M. Haynes (Haynes [6]).

The software was developed in-house, using the Lahey 80386 FORTRAN 77 compiler. Developed programs include:

Kessler Model Flux computation
Kessler Model Velocity and Direction Distribution code
Monte Carlo Collision Probability code
Analytic Collision Probability code
Maneuver Rate Determination code

Maneuver Ellipsoid code
Monte Carlo Microgravity Period code

Late in the analysis process, the Analytic Collision Probability code was developed and used for the runs with perfect tracking accuracy for the space vehicle. The main analysis effort consisted of a series of 10,000 Monte Carlo runs used to compute collision probabilities for the parameterizations described in the previous sections.

The Kessler Model Flux computation is outlined in appendix A as is the Kessler Model Velocity and Direction Distribution calculation. The mathematics of the Monte Carlo Collision Probability and Analytic Collision Probability code are shown in appendices B and C; detailed explanations and outlines of these methods are presented in the next section. An explanation and outline of the Maneuver Rate Determination code, used to compute the maneuver rate as a function of tracking accuracy and hit probability, is also presented in the next section with the mathematical details given in appendix D. The mathematics of the tool that calculates spatial dimensions associated with hit probability and tracking accuracies (Maneuver Ellipsoid code) is given in appendix E. Finally, the Microgravity Period code, which uses the anticipated maneuver rate to compute the probability and number of maneuver-free periods of specified length per year based on an anticipated maneuver rate, is examined in appendix F.

3.3 Methods

To understand the operational impact of orbital debris, a method was devised to relate the debris environment flux and the state vector uncertainties (for space vehicle and debris object) to the expected maneuver rate and to reduction in collision probability. Essentially, the process computes hit probabilities, the expected maneuver rate, assuming a maneuver is executed at a selected risk level, and the reduction to total collision probability obtained by maneuvering. The key algorithms, those for the single conjunction Monte Carlo Collision Probability, the Analytic Collision Probability calculations, and the Maneuver Rate Determination, are outlined in sections 3.3.1 through 3.3.3.

Before the details are presented, an introduction of some terms and concepts is warranted. Both the Monte Carlo and analytic methods of collision probability determination involve placing the debris object at the estimated closest point of approach (cpa) relative to the space vehicle. The uncertainties for the debris and space vehicle state vectors determine the spatial statistical distribution of the cpa with respect to the space vehicle. The Monte Carlo method samples the cpa statistical distribution while the analytic method assumes the mathematical form of the cpa distribution. Collision probability is defined as the probability of the cpa being within a 60-meter radius sphere about the space vehicle; this is a single event (conjunction) collision probability, P_e. The

60-meter sphere completely encompasses a space station-sized vehicle. For the range of uncertainties listed in sections 2.1 and 2.2, collision probability will be essentially constant throughout this sphere. A particular P_e , called P_m , is chosen so that the space vehicle always maneuvers if $P_e \geq P_m$ and doesn't maneuver otherwise. This condition ($P_e \geq P_m$) combined with the debris flux yields maneuver rate and the number of collisions avoided over time. These avoided collisions are a fraction of the collisions that could have occurred if no maneuvers are executed; therefore, the term fractional risk reduction refers to the factor by which collisions are reduced by maneuvering. The complement of the fractional risk reduction is the fractional residual risk. The fractional residual risk is that fraction of the probability of collision over time, assuming no avoidance maneuvers, which remains if maneuvers are performed whenever $P_e \geq P_m$. Relating P_m with a unit flux gives a relative maneuver rate. Fractional residual risk and relative maneuver rate are both independent of debris flux and vehicle size.

3.3.1 *Monte Carlo Collision Probability Code.* This section outlines the steps in the Monte Carlo method used to determine the collision probability for a single conjunction (event). An overview of the method is given below:

1. The debris object is placed at the estimated closest point of approach (cpa) relative to the space vehicle. This placement corresponds to the conjunction cpa predicted by USSPACECOM for an actual conjunction.

2. The positions of the space vehicle and the debris object are randomly displaced using a random number generator and the σ's from the position uncertainties, assuming normally distributed position errors.

3. A new cpa from the displaced positions is determined and a collision is said to occur if the cpa is within the 60-meter space vehicle sphere.

4. The process is repeated for one million trials.

5. The per-event collision probability, P_e , is calculated as
   P_e = (number of collisions)/(number of trials).

6. P_e may be normalized to any vehicle area.

The random displacements are never more than about 20 km. By placing the debris object at cpa relative to the space vehicle, the velocity errors, which don't exceed 1 m/sec, are not significant and the trajectories can be modeled as straight lines without introducing miss distance errors of more than a few centimeters.

3.3.2 *Analytic Method.* In the analytic method, the following process is used:

1. The position distribution of the debris object relative to the space vehicle is described by a two-dimensional, or bivariate, normal distribution (Hogg [7]) about the estimated cpa in a plane normal to the velocity vector of the debris object relative to the space vehicle.

2. The σ's associated with the bivariate normal distribution are calculated from the σ's of the space vehicle and the debris object.

3. A per-event collision probability, P_e, is calculated for the space vehicle sphere.

4. P_e can be normalized to an actual vehicle area even if the area varies with angle of incidence.

The mathematical justification for representing the position distribution by a bivariate normal distribution is discussed in appendix C. The σ's for the bivariate distribution can be computed from the Monte Carlo runs. The σ's from the Monte Carlo method agree with those calculated in step 2 above to 3 significant figures. The Monte Carlo and analytic methods give the same P_e values and yet are totally independent methods; thus, validation is provided for the bivariate distribution method.

3.3.3 *Maneuver Rate Determination Tool.* For maneuver rate determination, the following procedure is used:

1. Using the methods described above, a per-event collision probability, P_e, is determined for any space vehicle/orbital debris conjunction.

2. A specific per-event collision probability of P_m is selected such that the space vehicle always maneuvers if $P_e \geq P_m$ and never maneuvers otherwise.

3. Assume that the debris flux is from a single direction.

    a. Then the loci of points for constant P_e form concentric ellipses about the estimated cpa in a plane perpendicular to the relative velocity vector.

    b. There is an ellipse corresponding to $P_e = P_m$. The area of this ellipse times the debris flux is the maneuver rate.

    c. The area of this ellipse times the mean P_e over the ellipse times the annual flux is the number of collisions which would have occurred had the space vehicle not maneuvered (collisions avoided).

d. Assuming no debris avoidance maneuvers are executed, the annual collision probability is the area of the space vehicle times the annual flux.
e. The collisions avoided divided by the annual no-maneuver collision probability is the fractional risk reduction associated with the chosen P_m for the space vehicle and the UVW uncertainties. Fractional residual risk is one minus the fractional risk reduction. The fractional residual risk is then that fraction of the no-maneuver risk which remains when maneuvers are performed.

4. Generalizing from the single-direction results detailed above, relative maneuver rate and fractional residual risk can be obtained by integrating over the debris angle using the coefficients generated in appendix A. Although P_m is for a 60-meter radius sphere, the fractional risk reduction is independent of space vehicle size. This is because the collision probability within the sphere is nearly constant. The maneuver rate is determined by the P_m .

5. Actual collision probability over time for a specific space vehicle is determined by multiplying fractional risk reduction by space vehicle area and flux.

6. The relative maneuver rate is defined as the maneuver rate for a unit debris flux. The true maneuver rate is obtained by multiplying the relative maneuver rate by the actual flux.

4. Results

The general result of this study is that the fractional residual risk as a function of relative maneuver rate is independent of both debris flux and vehicle size. In this section the general result is discussed and then applied to a space station-sized vehicle.

Figure 2 shows fractional residual risk as a function of the relative maneuver rate for unit debris flux in km^{-2} $year^{-1}$, for P_m values 10^{-3}, 5×10^{-4}, 10^{-4}, 5×10^{-5}, 10^{-5}. Four groups of lines are presented corresponding to the four debris conjunction state vector error ellipsoids. The circles refer to the smallest debris error ellipsoids, with the diamonds, squares, and triangles referring respectively to increasingly larger debris error ellipsoids. The solid lines refer to the perfect space vehicle state vectors, with long dashed, medium dashed, and short dashed lines referring respectively to increasingly larger space vehicle error ellipsoids. The lines labeled "P_m" are the collision probability levels at which a debris avoidance maneuver is executed.

From this figure, two observations are made. First, for all state vector accuracies, the relative maneuver rate increases as fractional residual risk decreases; hence, as P_m

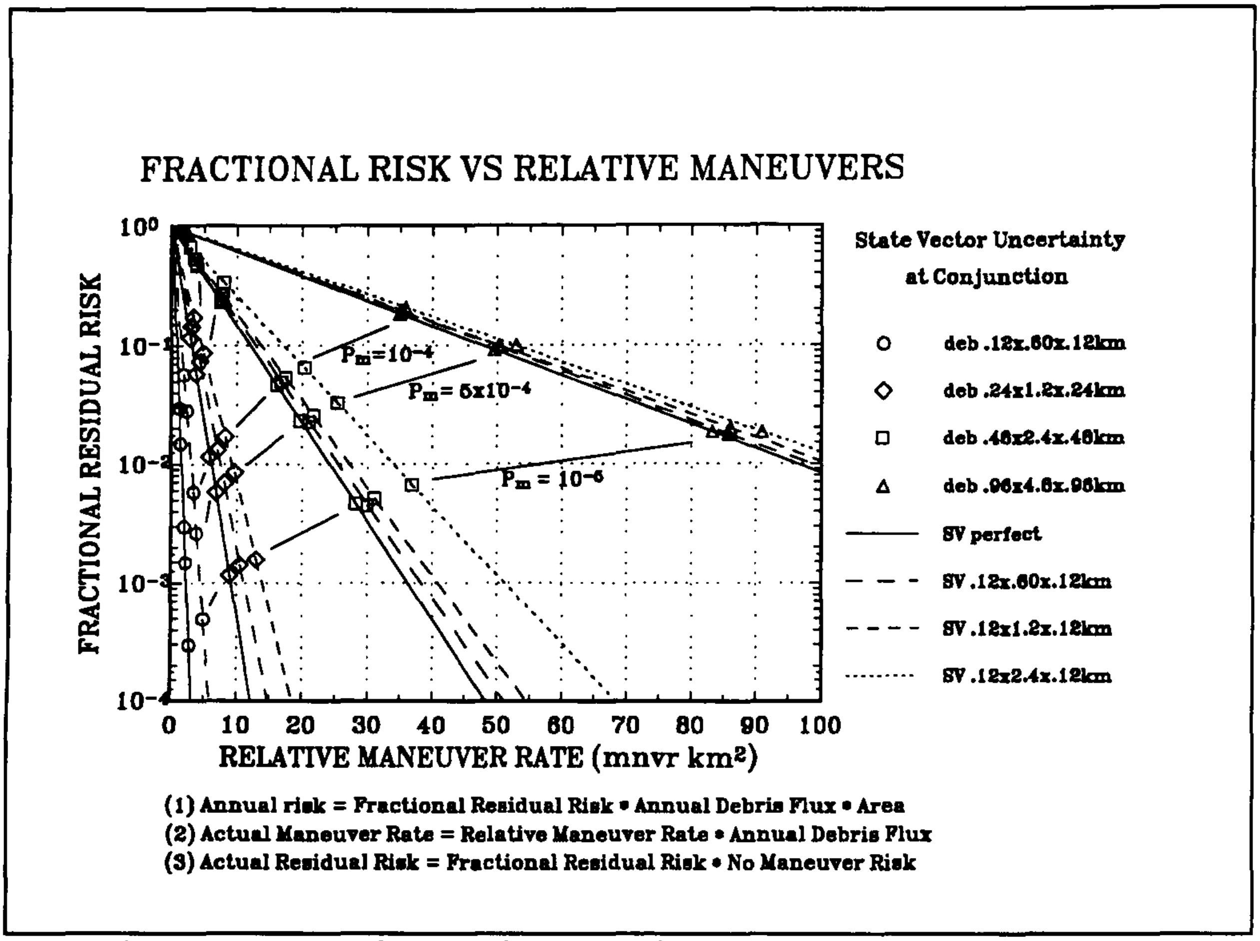

Figure 2. Fractional residual risk as a function of relative maneuver rate.

decreases, less probability of collision is accepted and more maneuvers are required. Second, given a relative maneuver rate, the P_m level is a function of space vehicle and debris state vector uncertainties. Example 1, on the next page, considers a specific space vehicle.

A space vehicle is to begin orbital operations in 1997 and continue in operation for at least 30 years. The vehicle has a cross-sectional area of 2000 m^2 viewed from any direction. State vectors for the vehicle are expected to have position uncertainties of 0.12 x 0.60 x 0.12 km (UVW). Debris state vectors are expected to have uncertainties of 0.12 x 0.60 x 0.12 km (UVW). The orbital debris population is expected to grow at a 5% annual rate over the next 30 years. In 1997, the TRACKED orbital debris flux is estimated to be 1.6 yr^{-1} km^{-2}. The program has decided to maneuver the vehicle whenever $P_c > 1 \times 10^{-5}$.

The expected 1997 debris avoidance maneuver rate for the space vehicle is taken from figure 2. Reading the long dashes/circle symbol line,

Relative Maneuver Rate = 5 mnvr km^2.

Using equation 2 from figure 2:

Actual Maneuver Rate = Relative Maneuver Rate * Annual Debris Flux,

Actual Maneuver Rate = (5 mnvr km^2) * (1.6 km^{-2} yr^{-1}) = 8 mnvr/yr.

The 1997 residual risk to the space vehicle if it performs all necessary collision avoidance maneuvers for TRACKED debris is also extracted from figure 2. Reading the long dashes/circle symbol line,

Fractional Residual Risk = 5×10^{-4}.

The 1997 annual risk of collision with TRACKED orbit debris for the space vehicle is taken from figure 2. Using equation 1 from figure 2:

Annual Risk = Fractional Residual Risk * Annual Debris Flux * Area,

Annual Risk = (5×10^{-4}) * (1.6 km^{-2} yr^{-1}) * 2000 m^2 * (1 km^2 / 1000^2 m^2)

Annual Risk = 1.6×10^{-6}

The lifetime risk of collision with ALL debris larger than 10 cm for the space vehicle, assuming no collision avoidance maneuvers, is determined using figure 1. Reading the triangle symbol line from figure 1:

Cumulative Collision Probability (for all objects > 10 cm) = 0.46.

The lifetime risk of collision with TRACKED debris larger than 10 cm for the space vehicle assuming no collision avoidance maneuver is determined using the equation from figure 1:

Cumulative Collision Probability for tracked objects larger than 10 cm =
Cumulative Collision Probability / 2.8 = 0.46 / 2.8 = 0.17.

The lifetime actual residual risk with avoidance maneuvers for TRACKED debris for the space vehicle is taken from figure 2 using equation 3:

Actual Residual Risk with avoidance maneuvers =
Fractional Residual Risk * Actual Risk (without maneuvers),

Actual Residual Risk = (5×10^{-4}) * (0.17) = 0.9×10^{-4}

Example 1. An example using figures 1 and 2.

If the particular vehicle considered in example 1 executes no collision avoidance maneuvers, it has a 46% probability of being hit by a debris object larger than 10 cm over a 30-year period. If no collision avoidance maneuvers are performed, the vehicle has a 17% probability of being hit by a TRACKED debris object over a 30-year period. With the given tracking accuracies, if a debris avoidance maneuver is executed whenever $P_e \geq 10^{-5}$, the fractional residual risk is 5×10^{-4}. In other words, the selected P_m criteria reduces the debris collision probability by a factor of 5×10^{-4}. In 1997, maneuvering when $P_e \geq 10^{-5}$, the anticipated maneuver rate for the vehicle will be 8 maneuvers/year while the probability of being hit by a tracked debris object will be 1.6×10^{-6}. If the vehicle maneuvers whenever $P_e \geq 10^{-5}$, the cumulative probability of the vehicle colliding with a tracked debris object over it's 30-year lifetime, is 0.9×10^{-4}.

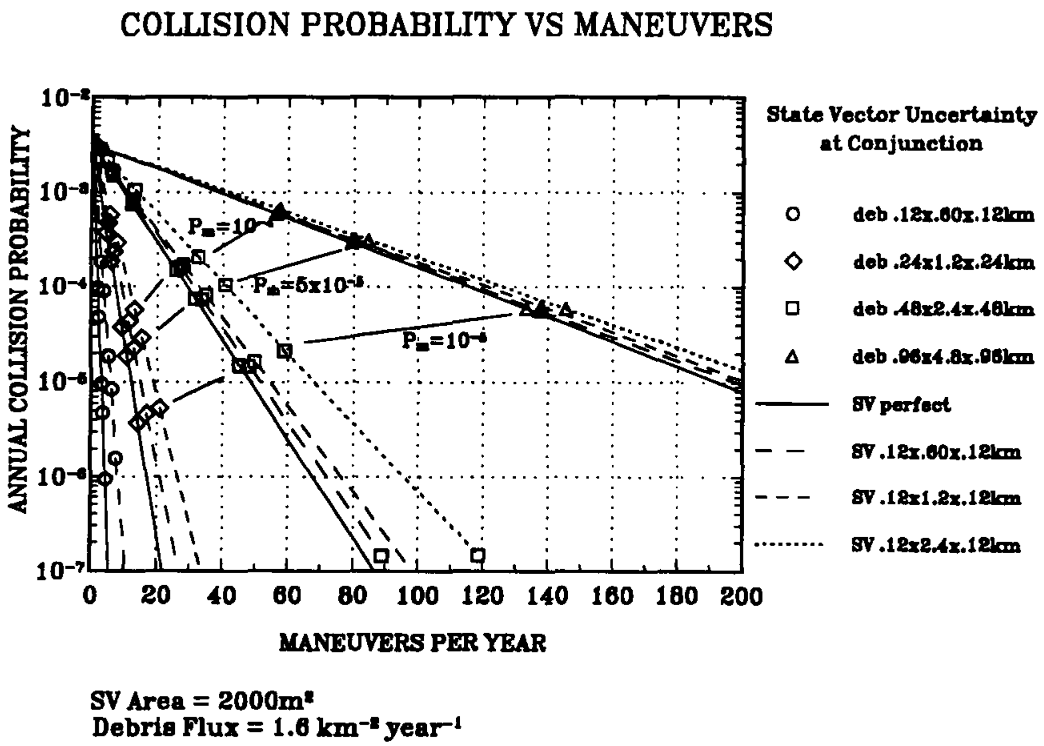

Figure 3. Annual collision probability as a function of annual maneuver rate.

Figure 3 shows the collision probability for a 2000m^2 cross-section space vehicle with a total debris flux of 1.6 km^{-2} year^{-1}. From figure 3, if no maneuvers are performed, the annual probability of collision with tracked debris is about 3.2 x 10^{-3}, independent of state vector accuracy. This annual collision probability is equal to the angle integrated flux, 1.6 km^{-2} year^{-1}, times the space vehicle cross-section of 2000 m^2, as it must be.

Figure 4 is an expansion of the low maneuver end of figure 3. Figures 3 and 4 are an instantiation of figure 2 for the example 1 vehicle, and correspond approximately to a completed space station with a low solar activity debris flux.

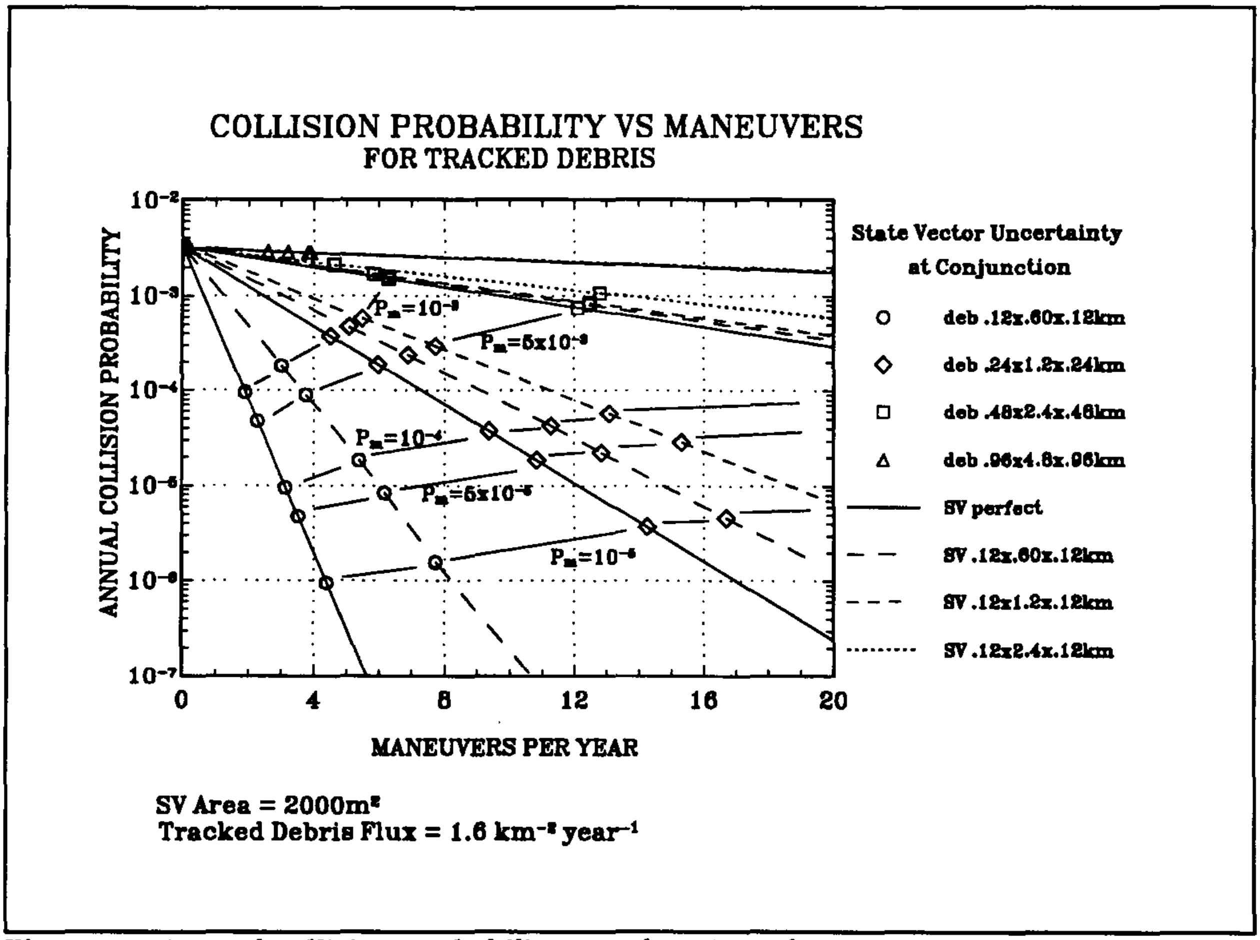

Figure 4. Annual collision probability as a function of maneuver rate.

Some important general ideas are apparent from figures 3 and 4.

a) The annual probability of collision varies with the state vector uncertainty ellipsoid sizes for both the space vehicle and the debris object and with the number of maneuvers performed, based on the P_m value.

b) For all state vector accuracies, the annual probability of collision decreases as the maneuver rate increases.

c) With better navigational accuracies, the number of conjunctions for a given P_m is lower. Therefore, a higher level of per conjunction collision probability can be accepted without increasing risk over time.

d) As the tracking accuracy on the debris state vector at conjunction improves, the number of conjunctions for all levels of collision probability is lessened. For constant P_m, both the maneuver rate and annual probability of collision are dramatically reduced.

e) For the smaller debris state vector uncertainties, slight increases in the maneuver rate correspond to very large reductions in annual probability of collision.

f) For good debris state vector accuracy, significant improvement in space vehicle navigation accuracy results in significant improvement in maneuver rate.

Examples 2 and 3 below demonstrate applications of figures 3 and 4.

For the same vehicle, state vector uncertainties for vehicle and debris, maneuver threshold (P_m), and debris flux as in example 1, the expected 1997 collision avoidance maneuver rate with tracked orbital debris can be taken from the line with long dashes and circles in figure 4:

Maneuvers Per Year $\approx$ 7.5.

This agrees with example 1 for which Maneuvers Per Year $\approx$ 8.

The 1997 annual risk of collision with tracked orbit debris can be read from figure 4. Looking at the line with long dashes and circles:

Annual Collision Probability $\approx 1.5 \times 10^{-6}$,

in agreement with the approximately 1.6×10^{-6} from example 1.

Example 2. An example using figure 4.

A certain earth orbiting space vehicle has a 2000 m^2 cross section as viewed from any direction. For a certain year this vehicle is expected to be exposed to a debris flux of 1.6 km^{-2} y^{-1}. The space vehicle UVW position uncertainty, propagated to conjunction, is expected to be 0.12 x 0.6 x 0.12 km, 1σ for every conjunction. The debris position accuracy, propagated to conjunction, is expected to be 0.12 x 0.6 x 0.12 km for 40% of the predicted conjunctions, 0.24 x 1.2 x 0.24 km for 30% of the conjunctions, 0.48 x 2.4 x 0.48 km for 20% of the conjunctions and 0.96 x 4.8 x 0.96 for the remaining 10% of the conjunctions.

Suppose the vehicle program adopts a P_m of 1×10^{-4}. From figure 3 we see that there would be 58 maneuvers per year with a collision probability of 6×10^{-4} if all debris had an uncertainty of 0.96 x 4.8 x 0.96 km and 28 maneuvers per year with a collision probability of 1.2×10^{-4} if all debris has an uncertainty of 0.48 x 2.4 x 0.48 km. From figure 4 we have maneuver rates of 11 and 5.5 maneuvers per year and collision probabilities of 4×10^{-5} and 2×10^{-5} associated with uncertainties of 0.24 x 1.2 x 0.24 km and 0.12 x 0.6 x 0.12 km. If no maneuvers are performed, the risk is 3.2×10^{-3}. The actual anticipated number of maneuvers per year is the sum of the number of maneuvers associated with each uncertainty times the fraction of conjunctions having that uncertainty. The anticipated annual collision probability is the sum of the collision probability associated with each uncertainty times the fraction of conjunctions having that uncertainty. The fractional residual risk is risk with maneuvers divided by risk without maneuvers.

Deb. Uncertainty	0.12x0.6x0.12	0.24x1.2x0.24	0.48x2.4x0.48	0.96x4.8x0.96
No. of Maneuvers	5.5	11	28	55
Coll. Prob.	2.0×10^{-5}	4.0×10^{-5}	1.5×10^{-4}	6.0×10^{-4}

Number of Maneuvers = 0.1*58 + 0.2*28 + 0.3*11 + 0.4*5.5 = 17.9
Ann. Collision Prob. = $0.1*6x10^{-4} + 0.2*1.5x10^{-4} + 0.3*4x10^{-5} + 0.4*2x10^{-5} = 1.1x10^{-4}$
Fractional Residual Risk = $1.1x10^{-4}/3.2x10^{-3} = 0.03$.

Thus, 97% of the risk is eliminated, while 3% remains.

Suppose the vehicle program adopts a P_m of $5x10^{-5}$. Then,

Deb. Uncertainty	0.12x0.6x0.12	0.24x1.2x0.24	0.48x2.4x0.48	0.96x4.8x0.96
No. of Maneuvers	3.8	6.8	12.5	3.9
Coll. Prob.	9.0×10^{-5}	2.3×10^{-4}	7.0×10^{-4}	2.8×10^{-3}

The anticipated number of maneuvers per year is
0.1*3.8 + 0.2*6.8 + 0.3*12.5 + 0.4*3.8 = 7.0

The number of collisions anticipated per year is
$0.1*2.8 \times 10^{-3} + 0.2*7 \times 10^{-4} + 0.3*2.3 \times 10^{-3} + 0.4*9 \times 10^{-5} = 5.3 \times 10^{-4}$

Fractional Residual Risk = $5.3 \times 10^{-4}/3.2 \times 10^{-3} = 0.16$.

Thus, 84% of the risk is eliminated while 16% remains.

Example 3. A hypothetical example using figures 3 and 4.

Figures 3 and 4 indicate additional important results. These figures show clearly that the probability of collision per year, the predicted state vector accuracies, and the avoidance maneuver rate for any space vehicle in low earth orbit are fundamentally interrelated. The relationship between these three parameters is such that once any two are specified, the third will be determined. Specifically:

Given an expected maneuver rate per year for any vehicle and the maximum acceptable probability per year of collision with debris objects, required state vector accuracies for both the vehicle and the debris object are determined.

Given the maximum acceptable probability per year of collision with debris objects and the accuracies of the vehicle and debris object state vectors, the expected debris avoidance maneuver rate for the vehicle is determined.

Given the state vector accuracies for the vehicle and debris objects and a maximum maneuver rate per year for the vehicle, the probability per year for a collision with debris objects is determined.

5. Conclusions

Better state vector or navigation accuracy for both debris and space vehicle decreases the maneuver rate for a given level of per-conjunction collision probability (P_m) and lowers the probability of collision over time. All space vehicle programs must consider the interrelationship between these three parameters when formulating avoidance strategies. Once any two are specified, the third is determined.

Knowledge of the debris state vector uncertainties is needed to develop estimates of probability of collision. Each debris object has a unique trajectory and is observed, with a particular tracking geometry, by a combination of sensors. Thus, there is a heterogenous set of debris object uncertainty ellipsoids, with variations in size and U-V-W axis ratios. The analyses to date have been performed to parameterize the probability of collision as a function of debris error ellipsoid size. With additional information on the distribution of debris ellipsoid sizes and geometries, the probability of collision can be refined, better understood, and managed.

The effectiveness of any candidate debris avoidance strategy should be measured by the annual fractional risk reduction it provides, independent of a particular debris flux or space vehicle size. Fractional residual risk can be evaluated for periods of time over which vehicle configuration is changing, without considering collision probability. Further, any debris avoidance strategy which does not result in a substantial annual risk reduction should not be adopted.

To successfully formulate a debris avoidance strategy, any program must specify two of the following three parameters: state vector accuracy, maneuver rate, and/or acceptable level of risk.

6. References

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Appendix A
Orbital Debris Flux Model

Flux Computation

An orbital debris flux model has been developed by Kessler et al (Kessler[1]).

This model gives debris flux, F_0, as a function of object size, d; altitude, h; time (date), t; and F10.7 solar flux, value S as:

$$F_0(d,h,i,t,S) = 4H(d)\phi(h,S)\psi(i)(F_1(d)g_1(t) + F_2(d)g_2(t))x10^{-6}\frac{impacts}{year\ Km^2}, \quad \text{(A1)}$$

upon a sphere, over the cross sectional area of the sphere rather than its surface.

d = lower limit debris size (cm).
t = decimal date as for example, 1998.32 = April 26,1998.
h = orbital altitude in km.
S = 13 month *smoothed* solar flux for (t-1)th year in 10_4 Jy
g_1 = fragment mass growth factor (2% until 2010, 4% after)
g_2 = mass factor (5% nominal value)
H = Henize factor, the estimate of unobserved objects with Size $\geq$ d
$\phi(h,S)$ = altitude distribution and solar flux attenuation factor
$\psi(i)$ = orbital inclination factor
$F_1(d)$, $F_2(d)$ = base object and fragment fluxes

The details and specific numeric values are found in reference 1.

Velocity and Direction Distribution

The debris flux model assumes all debris is moving with the same velocity, v, in a circular orbit. In this work the projectile will be the debris object and the target will be the space vehicle which will be in a circular orbit. Then all collisions will occur in a plane parallel to the earth's surface. From figure A-I

$$\vec{v}_r = \vec{v}_d - \vec{v}_{sv} \quad , \quad 2\theta_r + \theta = \pi \qquad \text{(A2)}$$

where θ = angle between orbits and θ_r is the relative angle the debris object makes with respect to the target object; $\vec{v}_s$ is the space vehicle velocity, $\vec{v}_d$ is the debris velocity and $\vec{v}_r$ is the relative velocity. From Figure A-II, the magnitude of $\vec{v}_r$ is given by

$$2v\cos\theta_r = v_r. \tag{A3}$$

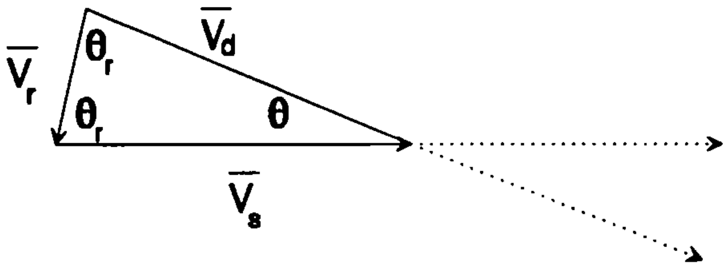

Figure A-I. Vector relation between SV, debris and relative velocities.

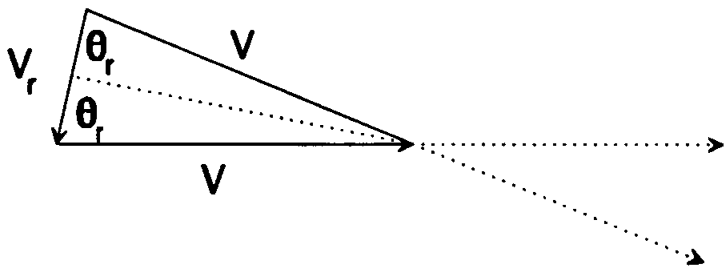

Figure A-II. Figure showing the relation 2v cos Θ = v_r.

The *unnormalized* flux distribution is given as a function of relative velocity in the debris flux model as

$$f(v_r)=(2v_r v_0-v_r^2)\{G(i)e^{-\frac{(v_r-A(i)v_0)^2}{(B(i)v_0)^2}}+F(i)e^{-\frac{(v_r-D(i)v_0)^2}{(E(i)v_0)^2}}\}+H(i)C(i)(4v_r v_0-v_0^2) \tag{A4}$$

where $A, B, C, D, E, F, G, H,$ and v_0 are functions of target object orbital inclination.

This can be transformed into a normalized function of relative angle θ_r, as

$$f(\theta_r)=\frac{f(v_r)\dfrac{dv_r}{d\theta_r}}{\displaystyle\int_{-\pi/2}^{+\pi/2} f(v_r)\frac{dv_r}{d\theta_r}d\theta_r}. \qquad \text{(A5)}$$

The annual flux per radian at relative angle θ_r, $F(\theta_r)$, is given by $F_0 f(\theta_r)$, where F_0 is calculated from equation A1. Integrating $f(\theta_r)$ over 15° intervals in relative angle, at the relative angles below we obtain $f(\theta_{ri})$ as

$$f(\theta_{ri})=\int_{\theta_i-7.5^\circ}^{\theta_i+7.5^\circ} f(\theta_r)d\theta_r\ . \qquad \text{(A6)}$$

The values of $f(\theta_{ri})$ obtained are shown in Table A-I below.

TABLE A-I.- $f(\theta_{ri})$ VALUES

θ_{ri}	15°	30°	45°	60°	75°
$f(\theta_{ri})$	.009	.295	.335	.289	.069

A region of ±7.5° about 0° and regions of 7.5° from ±90° relative angles are not included in table A-I and are excluded from the analysis. However, adding the $f(\theta_{ri})$ values in table A-I, we obtain 0.997; only 0.3% of the flux is not considered in our analysis. The amount of flux in a 15° interval about $\pm\theta_{ri}$ is given by $F_0 f(\theta_{ri})$.

Appendix B
Monte Carlo Determination of a Collision Probability for Two Orbiting Objects

Concept Definition

A collision between two orbiting objects is defined to occur if they pass within some distance R of one another. Given the state vector for each object at the estimated closest point of approach (cpa) and the uncorrelated normally distributed UVW position uncertainties for each object, the collision can be simulated. This is done by initially placing the two objects at the estimated cpa; then using a random number generator to produce normally distributed position uncertainties. The perturbed states are then propagated until a new point of closest approach is reached. If this cpa is within a distance R, then a simulated collision has occurred. Repeating this procedure a large number of times and dividing the number of collisions by the number of simulations, gives a probability. The velocity uncertainties do not enter into the simulation because the difference between projected and actual cpa is typically very small, 1 second or less, and the velocity uncertainties are on the order of 1 m/s.

Let $\bar{r}_s$ and $\bar{v}_s$ denote the position and velocity vectors associated with the space vehicle, space vehicle, and $\bar{r}_d$ and $\bar{v}_d$ denote the position and velocity vectors of the projectile, or debris object. The UVW uncertainty vectors associated with the space vehicle and the debris are written as $\bar{e}_s$ and $\bar{e}_d$. At the estimated point of closest approach let

$$\vec{r}_s = \vec{r}_{s0} \quad \text{and} \quad \vec{r}_d = \vec{r}_{d0} \; . \tag{B1}$$

Assume the space vehicle and projectile travel in straight lines and let t = 0 at estimated cpa. Then near the estimated point of closest approach the motion of both is described by

$$\vec{r}_s = \vec{r}_{s0} + \vec{v}_s\, t \quad \text{and} \quad \vec{r}_d = \vec{r}_{d0} + \vec{v}_d\, t \; . \tag{B2}$$

Including position uncertainties, perturbed positions are given by

$$\begin{gathered} \vec{r}_s = \vec{\tilde{r}}_{s0} + \vec{v}_s\, t \quad \text{and} \quad \vec{r}_d = + \vec{\tilde{r}}_{d0} + \vec{v}_d\, t \quad \text{where} \\ \vec{\tilde{r}}_{s0} = \vec{r}_{s0} + R_s \vec{e}_s \quad \text{and} \quad \vec{\tilde{r}}_{d0} = \vec{r}_{d0} + R_d \vec{e}_d \; . \end{gathered} \tag{B3}$$

The ~ indicates the position vectors perturbed by their uncertainties. R_s and R_d are the matrices rotating $\bar{e}_s$ and $\bar{e}_d$ into the reference frame of $\bar{r}_s$ and $\bar{r}_d$.

The distance between the space vehicle and debris object is given by

$$\vec{r}^{\sim} = \vec{r}_{d0} - \vec{r}_{s0} + R_d\vec{e}_d - R_s\vec{e}_s + (\vec{v}_d - \vec{v}_s)t \ . \tag{B4}$$

Then

$$\begin{gathered}\vec{r}^{\sim} = \vec{r}_0 + R_d\vec{e}_d - R_s\vec{e}_s + \vec{v}_r\, t = \vec{r}_0^{\sim} + \vec{v}_r t \quad \text{where} \\ \vec{r}_0 = \vec{r}_{d0} - \vec{r}_{s0}\ , \quad \vec{v}_r = \vec{v}_d - \vec{v}_s\ , \quad \text{and} \quad \vec{r}_0^{\sim} = \vec{r}_{d0}^{\sim} - \vec{r}_{s0}^{\sim}\ .\end{gathered} \tag{B5}$$

The Monte Carlo method involves placing space vehicle and debris object in the cpa geometry, calling the FORTRAN supplied random number generator obtaining pairs (u_1,u_2) of unifromly distributed (0,1) random numbers, and using the Box Mueller Transformation,

$$x_1 = \sqrt{-2\ln(u_1)}\ \cos(2\pi u_2) \quad \text{and} \quad x_2 = \sqrt{-2\ln(u_1)}\ \sin(2\pi u_2)\ , \tag{B6}$$

to generate pairs (x_1,x_2) of uncorrelated normally distributed uncertainties.

The geometry of time of closest approach must satisfy $d/dt\{\vec{r}^{\sim} \cdot \vec{r}^{\sim}\} = 0$. That is

$$\begin{aligned}\frac{d}{dt}\{\vec{r}^{\sim}\cdot\vec{r}^{\sim}\} &= \frac{d}{dt}\{(\vec{r}_0^{\sim}+\vec{v}_r t)\cdot(\vec{r}_0^{\sim}+\vec{v}_r t)\} = \frac{d}{dt}\{\vec{r}_0^{\sim}\cdot\vec{r}_0^{\sim}+2(\vec{r}_0^{\sim}\cdot\vec{v}_r)t+(\vec{v}_r\cdot\vec{v}_r)t\} \\ &= 2\vec{r}_0^{\sim}\cdot\vec{v}_r + 2(\vec{v}_r\cdot\vec{v}_r)t = 0\ .\end{aligned} \tag{B7}$$

Then solving equation B7 and substituting from equation B4, time of cpa is

$$\begin{gathered}t_{cpa} = -(\vec{r}_0^{\sim}\cdot\vec{v}_r)/(\vec{v}_r\cdot\vec{v}_r) = \\ -\{(\vec{r}_0+R_d\vec{e}_d-R_s\vec{e}_s)\cdot\vec{v}_r\}/(\vec{v}_r\cdot\vec{v}_r)\ .\end{gathered} \tag{B8}$$

Clearly if $\vec{e}_s = 0$ and $\vec{e}_d = 0$, then $t_{cpa} = 0$, since $t \equiv 0$ at estimated cpa.

Thus $\vec{r}_0 \cdot \vec{v}_r = 0$, and the projected relative cpa vector, $\vec{r}_0$, is perpendicular to the relative velocity vector, $\vec{v}_r$. Then,

$$t_{cpa} = \{(R_d\vec{e}_d-R_s\vec{e}_s)\cdot\vec{v}_r\}/(\vec{v}_r\cdot\vec{v}_r)\ . \tag{B9}$$

Using equations B4 and B5 at $t = t_{cpa}$, the minimum miss distance for the "perturbed" state vectors is then

$$\vec{r}^{\sim} = \vec{r}_{d0} - \vec{r}_{s0} + R_d \vec{e}_d - R_s \vec{e}_s + (\vec{v}_d - \vec{v}_s) t_{cpa} \quad . \tag{B10}$$

This quantity is the basis in performing the Monte Carlo analysis. The probability of a miss distance r being $\leq$ some radius R is

$$P(r \leq R) = \frac{Number\ of\ occurances\ of\ r \leq R}{Number\ of\ Monte\ Carlo\ tries} \quad . \tag{B11}$$

Implementation

In the implementation of the above equations we make the assumptions inherent in the Kessler Model:

1. The velocity vectors of the two objects are in a plane parallel to the surface of the earth. Thus the U axes in the space vehicle and debris systems are parallel.
2. The magnitude of the velocities, v, is the same for the two objects.

We chose to work in the space vehicle UVW coordinate system, with unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ along the U,V, and W axes respectively and with the origin located at the geometric center of the space vehicle. Then,

$$\begin{gathered} \vec{v}_s = v\hat{j} \quad , \quad \vec{v}_d = v\cos(\theta)\hat{j} + v\sin(\theta)\hat{k} \quad \text{and} \\ \vec{v}_r = \vec{v}_d - \vec{v}_s = v(\cos(\theta)-1)\hat{j} + v\sin(\theta)\hat{k} \quad . \end{gathered} \tag{B12}$$

Also

$$\vec{r}_{s0} = 0 \quad \text{and} \quad \vec{r}_{s0}^{\sim} = \vec{r}_{s0} + \vec{e}_s = \vec{e}_s = e_{s1}\,\hat{i} + e_{s2}\,\hat{j} + e_{s3}\,\hat{k} \tag{B13}$$

and

$$\begin{gathered} \vec{r}_{d0} = r_{dx0}\,\hat{i} + r_{dy0}\,\hat{j} + r_{dz0}\,\hat{k} \quad \text{so that} \\ \vec{r}_{d0}^{\sim} = \vec{r}_{d0} + R_d \vec{e}_d = \vec{r}_{d0} + (e_{d2}\cos\theta + e_{d3}\sin\theta)\,\hat{j} + (-e_{d2}\sin\theta + e_{d3}\cos\theta)\,\hat{k} \end{gathered} \tag{B14}$$

with $r_{dx0} = d_{up}$, $r_{dy0} = -\,d_{hor}\sin(\theta_r)$, $r_{dz0} = d_{hor}\cos(\theta_r)$, where
v is the magnitude of the orbital velocities for the two objects, θ is the angle between their orbits and θ_r is angle of relative motion as seen from the space vehicle; d_{hor} and d_{up} are the estimated cpa distances in the horizontal (yz or vw) plane and the vertical (x or u) direction, in the space vehicle system.

Using equations 13 and 14

$$\overrightarrow{r_0^{\sim}} = \overrightarrow{r_{d0}^{\sim}} - \overrightarrow{r_{s0}^{\sim}} = (r_{dx0}+e_{d1}-e_{s1})\hat{i} + (r_{dy0}+e_{d2}\cos\theta+e_{d3}\sin\theta-e_{s2})\hat{j} + (r_{dz0}-e_{d2}\sin\theta+e_{d3}\cos\theta-e_{s3})\hat{k}\ . \qquad \textbf{(B15)}$$

From $\bar{v}_r = \bar{v}_d - \bar{v}_s$, using equations B12,

$$\bar{v}_r \cdot \bar{v}_r = v^2\{(\cos\theta - 1)^2 + \sin^2\theta\} = 2\ v^2(1\text{-}\cos\Theta) \qquad \textbf{(B16)}$$

where v is the magnitude of the orbital velocity. Then

$$t_{cpa} = -\ \{(\cos\theta - 1)(r_{dy0} + e_{d2}\cos\theta + e_{d3}\sin\theta - e_{s2}) + \sin\theta(r_{dz0} + e_{d2}\sin\theta - e_{d3}\cos\theta - e_{s3})\}/\{2v(1\text{-}\cos\theta)\}\ . \qquad \textbf{(B17)}$$

The perturbed miss distance vector becomes

$$\overrightarrow{r^{\sim}} = \overrightarrow{r_0^{\sim}} + v\{(\cos\theta-1)\hat{j} + \sin\theta\hat{k}\}t = \tilde{r}_x\ \hat{i} + \tilde{r}_y\ \hat{j} + \tilde{r}_z\ \hat{k} \qquad \textbf{(B18)}$$

with

$$\begin{aligned}
\tilde{r}_x &= r_{d1} + e_{d1} - e_{s1}\ , \\
\tilde{r}_y &= r_{d2} + e_{d2}\cos\theta + e_{d2} + e_{d3}\sin\theta - e_{s2} + v(\cos\theta - 1) \\
\tilde{r}_y &= r_{d3} + e_{d2}\sin\theta + e_{d3} + e_{d3}\cos\theta - e_{s3} + v\ \sin\theta\ t
\end{aligned} \qquad \textbf{(B19)}$$

and

$$Miss\ Distance = \sqrt{\tilde{r}_x^{\,2}+\tilde{r}_y^{\,2}+\tilde{r}_z^{\,2}}\ . \qquad \textbf{(B20)}$$

The Monte Carlo procedure generates a probability of a *miss distance* $\leq R$, where R is the the raduis of a sphere about the space vehicle. Each computed trial tests for a miss distance being less than 10 different values of R_0. The Monte Carlo analysis samples 10^6 trials for each unique geometry of debris object relative to space vehicle (cpa geometry).

Appendix C
Numeric Analytic Determination of a Collision Probability for Two Orbiting Objects

Define an orthogonal coordinate system in terms of debris and space vehicle velocity vectors, $\bar{v}_d$ and $\bar{v}_s$, and their difference, $\bar{v}_r = \bar{v}_d - \bar{v}_s$, as

$$\hat{U} = (\vec{v}_s \times \vec{v}_d) \,/\, |\vec{v}_s \times \vec{v}_d| \quad , \quad \hat{V} = (\vec{v}_r) \,/\, |\vec{v}_r| \quad , \quad \hat{W} = \hat{U} \times \hat{V} \ . \tag{C1}$$

In this section the rotation matrices rotate vectors into the *UVW* coordinate frame, whereas in appendix B, the rotations were into the space vehicle *UVW* frame.

In appendix B, equation B8, it was shown that $\bar{v}_r \cdot \bar{r}_0 = 0$ where $\bar{r}_0 = \bar{r}_{d0} - \bar{r}_{s0}$ is the estimated cpa vector and

$$t_{cpa} = -\,(\vec{r}_0^{\,\sim} \cdot \vec{v}_r) \,/\, (\vec{v}_r \cdot \vec{v}_r) \ . \tag{C2}$$

At t_{cpa} the quantity

$$(\vec{r}^{\,\sim} - \vec{r}_0) \cdot \vec{v}_r = (\vec{r}_0 + R_d \vec{e}_d - R_s \vec{e}_s + \vec{v}_r t - \vec{r}_0) \cdot \vec{v}_r \tag{C3}$$

becomes zero by substituting for $\vec{r}^{\,\sim}$ in equation 3 using equation 5 from appendix B:

$$\begin{gathered} (R_d \vec{e}_d) \cdot \vec{v}_r - (R_s \vec{e}_s) \cdot \vec{v}_r + (\vec{v}_r \cdot \vec{v}_r) t_{cpa} = \\ (R_c \vec{e}_d) \cdot \vec{v}_r - (R_s \vec{e}_s) \cdot \vec{v}_r - \vec{v}_r \cdot \vec{v}_r \{[(\vec{r}_0 + R_d \vec{e}_d - R_s \vec{e}_s) \cdot \vec{v}_r] \,/\, (\vec{v}_r \cdot \vec{v}_r)\} = 0 \ . \end{gathered} \tag{C4}$$

Thus, the problem is deterministic along the relative velocity vector since there is no position uncertainty in that direction.

Assume normally distributed UVW uncertainties for debris and space vehicle. Then the covariance matrix is defined as $E[(\bar{r}^{\sim} - \bar{r}_0)(\bar{r}^{\sim} - \bar{r}_0)^T]$, the expected value of $(\bar{r}^{\sim} - \bar{r}_0)(\bar{r}^{\sim} - \bar{r}_0)^T$.

At t_{cpa}, $E[(\bar{r}^{\sim}-\bar{r}_0)(\bar{r}^{\sim}-\bar{r}_0)^T]$ becomes

$$E[(\bar{r}^{\sim}-\bar{r}_0)(\bar{r}^{\sim}-\bar{r}_0)^T] = E[(\mathbf{R}_d\bar{\mathbf{e}}_d - \mathbf{R}_s\bar{\mathbf{e}}_s - \mathbf{v}_r t)(\mathbf{R}_d\bar{\mathbf{e}}_d - \mathbf{R}_s\bar{\mathbf{e}}_s - \bar{v}_r t)^T]$$

$$= E[(\mathbf{R}_d\bar{\mathbf{e}}_d - \mathbf{R}_s\bar{\mathbf{e}}_s)\,(\mathbf{R}_d\bar{\mathbf{e}}_d - \mathbf{R}_s\bar{\mathbf{e}}_s)_{UW}{}^T] \qquad \textbf{(C5)}$$

since $\bar{v}_r \cdot (R_d\bar{e}_d - R_s\bar{e}_s - \bar{v}_r t) = 0$.

The *UW* subscript means we take only $\hat{U}$ and $\hat{W}$ components. Then

$$E[\bar{r}^{\sim}-\bar{r}_0)(\bar{r}^{\sim}-\bar{r}_0)^T] = E[(R_d\bar{e}_d\ \bar{e}_d{}^T R_d{}^T)_{UW}] - E[(R_d\bar{e}_d\ \bar{e}_s{}^T R_s{}^T)_{UW}] - E[(R_s\bar{e}_s\ \bar{e}_d{}^T R_d{}^T)_{UW}]$$

$$+ E[(R_s\bar{e}_s\ \bar{e}_s{}^T R_s{}^T)_{UW}] \qquad \textbf{(C6)}$$

$$= E[(R_d\bar{e}_d\ \bar{e}_d{}^T R_d{}^T)_{UW}] + E[(R_s\bar{e}_s\ \bar{e}_s{}^T R_s{}^T)_{UW}] \ .$$

This is because

$$E[(R_d\bar{e}_d\ \bar{e}_s{}^T R_s{}^T)_{UW}] = E[(R_s\bar{e}_s\ \bar{e}_d{}^T R_d{}^T)_{UW}] = 0\ ,$$

since the components of $\bar{e}_d$ and $\bar{e}_s$ are all uncorrelated.

For debris and space vehicle moving parallel to the earth with the same velocity, the radial U component vectors are aligned and the resulting two-dimensional covariance matrix is diagonal.

Figure C-I shows the geometry and rotations involved in this restricted problem.

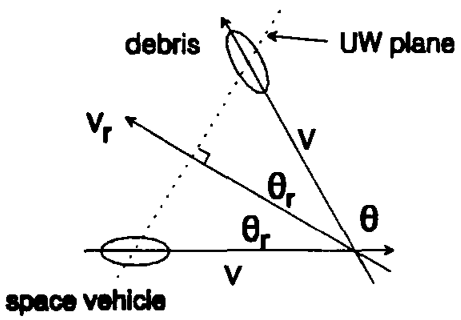

Figure C-1. Illustration of geometry and rotations in this restricted problem.

The σ_U and σ_W associated with the U and W directions are the related to the σ_{si} and σ_{di} associated with the space vehicle and debris as:

$$\begin{aligned} \sigma_U^2 &= \sigma_{s1}^2 + \sigma_{d1}^2 \\ \sigma_W^2 &= \sigma_{s2}^2\sin^2\theta_r + \sigma_{s3}^2\cos^2\theta_r + \sigma_{d2}^2\sin^2\theta_r + \sigma_{d3}^2\cos^2\theta_r \ ; \end{aligned} \tag{C7}$$

where

$$\begin{aligned} \sigma_{s1}^2 &= E[e_{s1}^2] \ , \ \sigma_{s2}^2 = E[e_{s2}^2] \ , \ \sigma_{s3}^2 = E[e_{s3}^2] \ , \\ \sigma_{d1}^2 &= E[e_{d1}^2] \ , \ \sigma_{d2} = E[e_{d2}^2] \ , \ \sigma_{d3} = E[e_{d3}^2] \ . \end{aligned} \tag{C8}$$

Thus the distribution of miss distance, $\bar{r}^{\sim} - \bar{r}_0$, is bivariate normal in the U-W plane. The density function in two dimensions is

$$f(UW) = \frac{1}{2\pi\sigma_U\sigma_V} e^{-\frac{(U-U_0)^2}{2\sigma_U^2}} e^{-\frac{(W-W_0)^2}{2\sigma_W^2}} \ , \tag{C9}$$

where U_0 and W_0 are projected miss distances along the U and W axes. Expressing U, W, U_0, and W_0 in polar coordinates, we obtain

$$U = r\sin\theta \quad , \quad W = r\cos\theta \quad , \quad U_0 = R_0\sin\phi \quad , \quad W_0 = R_0\cos\phi, \tag{C10}$$

where

$$R_0 = \sqrt{U_0^2 + W_0^2} \quad , \quad r = \sqrt{U^2 + W^2} \ .$$

R_0 and ϕ define the anticipated miss location, while r and θ define the actual spacial position. The probability of a miss distance $\leq$ to some radius R is

$$P(r \leq R) = \frac{1}{2\pi\sigma_U\sigma_W} \int_{r=0}^{R} \int_{\theta=0}^{2\pi} e^{-\frac{(r\cos\theta - R_0\cos\phi)^2}{2\sigma_W^2}} e^{-\frac{(r\sin\theta - R_0\sin\phi)^2}{2\sigma_U^2}} r\,dr\,d\theta \tag{C11}$$

$$= \frac{1}{2\pi\sigma_U\sigma_W} e^{-R_0^2\left(\frac{\cos^2\phi}{2\sigma_W^2} + \frac{\sin^2\phi}{2\sigma_U^2}\right)} \int_0^R \int_0^{2\pi} e^{+rR_0\left(\frac{\cos\phi\cos\theta}{\sigma_W^2} + \frac{\sin\phi\sin\theta}{\sigma_U^2}\right) - r^2\left(\frac{\sin^2\theta}{2\sigma_U^2} + \frac{\cos^2\theta}{2\sigma_W^2}\right)} r\,dr\,d\theta \ .$$

This is easily integrated numerically and runs in less than 1 sec on a 33MHz PC.

Appendix D
Maneuver Rate Determination Procedure

Understanding the maneuver rate determination procedure is central to an understanding of this work. This is the procedure by which the hit probability computation results were combined to give risk mitigation as a function of maneuver rate.

Unidirectional Flux

To demonstrate the method employed, we assume all debris comes from one horizontal direction. From the collision probability codes (appendices B and C), hit probability is calculated as a function of radial and horizonal miss distance, angle between orbits, for the chosen parameterizations of debris and space vehicle position uncertainty estimate at time of conjunction.

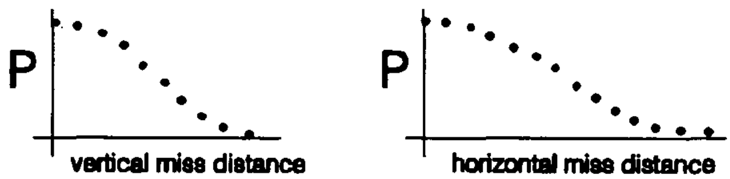

Figure D-I. Figure showing collision probability P as a function of vertical and horizontal miss distance.

Combining these two sets of data defines miss distance ellipses of constant hit probability as shown in figure D-II.

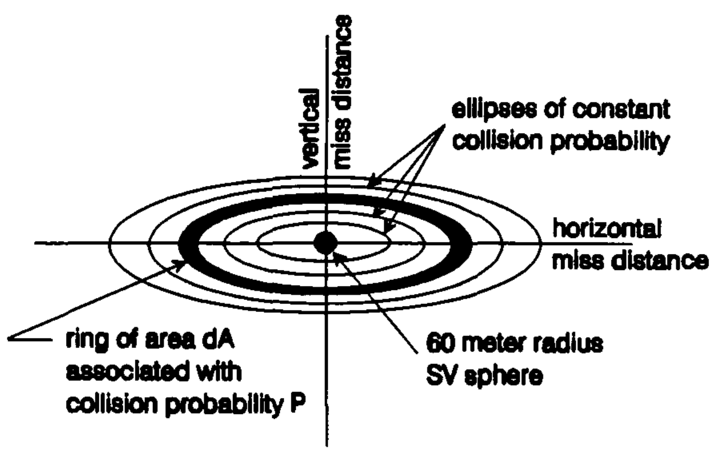

Figure D-II. Illustration of ellipses of constant collision probability as seen by unidirectional flux.

Let F be the component of annual debris flux making relative angle θ_r with respect to the space vehicle velocity vector. The flux component $F(\theta_r)$ is then perpendicular to a system of ellipses. Consider an elliptical ring with area dA and associated hit probability P. Then the product FdA is the annual number of debris objects in flux F passing through the ring in a year and the hit probability to the space vehicle sphere over a year is $PFdA$. We may write $P = P(\theta_r, A)$ where by A we refer to the probability per unit area associated with the elliptical ring rather than that associated with the entire ellipse; similarly F may be written to as $F(\theta_r)$. For mathematical convenience we define the no avoidance maneuver hit probability (or risk), P_T , to be the probability of a debris passage through a 60 meter radius sphere about the space vehicle. This is equal to the total number of passages per year through this sphere, due to flux F and is given by

$$P_T = FA_{\otimes} = \int_{A=0}^{\infty} PFdA \qquad \text{(D1)}$$

where $A_{\otimes}$ is the cross section of the space vehicle sphere. It is then evident that

$$A_{\otimes} = \int_{A=0}^{\infty} PdA \qquad \text{(D2)}$$

Suppose that P_m is the maximum acceptable hit probability to the 60-meter radius vehicle sphere for the combination of debris and station under consideration and let us require that the space vehicle maneuver if the hit probability P_e for the conjunction geometry equals or exceeds this value. Then

$$FA_{\otimes} = \int_{A=0}^{\infty} PFdA = \int_{A=0}^{A(Pm)} PFdA + \int_{A(Pm)}^{\infty} PFdA\ . \qquad \text{(D3)}$$

We may identify

$$Q = \int_{A=0}^{A(P_m)} PFdA \qquad \text{(D4)}$$

as risk reduction if the vehicle always maneuvers for $P_e \geq P_m$ and

$$R = \int_{A(P_m)}^{\infty} PFdA \tag{D5}$$

as remaining risk if the vehicle never maneuvers for $P_e < P_m$. Then total risk with no maneuvers, P_T , is related to Q and R as

$$P_T = Q + R \,. \tag{D6}$$

By dividing equation D3 through by $FA_\otimes$ and considering equation D2 we obtain

$$\frac{1}{A_\otimes}\int_0^\infty PdA = \frac{1}{A_\otimes}\int_0^{A(P_m)} PdA + \frac{1}{A_\otimes}\int_{A(P_m)}^{\infty} PdA = 1 \;. \tag{D7}$$

We interpret

$$Q_F = \frac{1}{A_\otimes}\int_0^{A(Pm)} PdA \quad \text{as fractional risk reduction} \tag{D8}$$

and

$$R_F = \frac{1}{A_\otimes}\int_{A(Pm)}^{\infty} PdA \quad \text{as fractional residual risk .} \tag{D9}$$

Then equation D7 states that

$$R_F + Q_F = 1 \,. \tag{D10}$$

Since the number of maneuvers per year, N_m, is simply the flux times the area of the ellipse defined by P_m and the flux from a given direction is constant, the number of maneuvers per year is given by

$$N_m = F\int_{A=0}^{A(Pm)} dA \quad . \tag{D11}$$

Multidirectional Flux

We now take into account the multidirectionality of debris flux in the horizontal plane. In order to compute a total collision probability, it is necessary to integrate debris flux over angle. From the debris flux model discussion in appendix A we may write

$$F(\theta_r) = F_0\, f(\theta_r) \quad \text{with} \quad \int_{-\pi/2}^{+\pi/2} f(\theta_r)d\theta_r = 1. \tag{D12}$$

where F_0 is the Kessler model flux from equation A1 (appendix A) and $f(\theta_r)$ is defined in equation A6.

The total collision probability equation, equation D1, becomes

$$P_T = A_{\otimes}F_0 \int_{-\pi/2}^{+\pi/2} f(\theta_r)d\theta_r = A_{\otimes}F_0 = F_0 \int_{-\pi/2}^{+\pi/2} \int_{A=0}^{\infty} P(A,\theta_r)f(\theta_r)dAd\theta_r\ . \tag{D13}$$

Risk reduction if a maneuver is always executed if $P_e \geq P_m$, equation D4, becomes

$$Q = F_0 \int_{-\pi/2}^{+\pi/2} \int_{A=0}^{A(P_m,\theta_r)} P(A,\theta_r)f(\theta_r)dAd\theta_r\ , \tag{D14}$$

and residual risk if no maneuver if $P \leq P_m$, equation D5 becomes

$$R = F_0 \int_{-\pi/2}^{+\pi/2} \int_{A(P_m,\theta_r)}^{\infty} P(A,\theta_r)\ f(\theta_r)dAd\theta_r\ . \tag{D15}$$

The number of maneuvers per year, equation D11, is now given by

$$N_m = F_0 \int_{-\pi/2}^{+\pi/2} \int_{A=0}^{A(P_m,\theta_r)} f(\theta_r)dAd\theta_r\ . \tag{D16}$$

Assuming a *spherical space vehicle* of cross-section $A_\otimes$ and recalling equation D2

$$F_0 \int_{-\pi/2}^{+\pi/2} \int_{A=0}^{\infty} P(A,\theta_r) f(\theta_r) dA d\theta_r = F_0 \int_{-\pi/2}^{+\pi/2} \{ \int_{A=0}^{\infty} P(A,\theta_r) dA \} f(\theta_r) d\theta_r$$
$$= F_0 A_\otimes \int_{-\pi/2}^{+\pi/2} f(\theta_r) d\theta_r = F_0 A_\otimes \quad . \qquad \text{(D17)}$$

Then, using equation D17, we obtain an equation analogous to equation D7,

$$\frac{1}{A_\otimes} \int_{-\pi/2}^{+\pi/2} \int_{0}^{\infty} P(A,\theta_r) f(\theta_r) dA d\theta_r = \frac{1}{A_\otimes} \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} P(A,\theta_r) f(\theta_r) dA d\theta_r$$
$$+ \frac{1}{A_\otimes} \int_{-\pi/2}^{+\pi/2} \int_{A(P_m,\theta_r)}^{\infty} P(A,\theta_r) f(\theta_r) dA d\theta_r = 1 \ , \qquad \text{(D18)}$$

from which we can interpret

$$Q_F = \frac{1}{A_\otimes} \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} P(A,\theta_r) f(\theta_r) dA d\theta_r \quad \text{as fractional risk reduction} \qquad \text{(D19)}$$

and

$$R_F = \frac{1}{A_\otimes} \int_{-\pi/2}^{+\pi/2} \int_{A(P_m,\theta_r)}^{\infty} P(A,\theta_r) dA \ f(\theta_r) d\theta_r \quad \text{as fractional residual risk} \ . \qquad \text{(D20)}$$

A relative maneuver rate is given by

$$N_{mR} = \frac{N_m}{F_0} = \int_{-\pi/2}^{+\pi/2} \int_{A=0}^{A(P_m,\theta_r)} f(\theta_r) dA d\theta_r \quad . \qquad \text{(D21)}$$

Figure 2 in the main text shows R_F on the vertical axis as "fractional residual risk" and N_{mR} on the x-axis as "relative maneuver rate."

Figures 3 and 4 in the main text show the actual residual risk, R (from equation D13) on the vertical axis and the actual expected number of maneuvers, N_m (from equation D14) on the horizontal axis. Residual risk, R, and fraction residual risk, R_F, are related as

$$Q = A_{\otimes} F_0 Q_F = A_{\otimes} F_0 \left\{ \frac{1}{A_{\otimes}} \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} P(A,\theta_r) f(\theta_r) dA d\theta_r \right\} \qquad \text{(D22)}$$

and actual maneuver rate is related to relative maneuver rate as

$$N_m = F_0 N_{mF} = F_0 \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} f(\theta_r) dA d\theta_r \quad . \qquad \text{(D23)}$$

These integrals are evaluated numerically as

$$F_0 \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} P(A,\theta_r)\, f(\theta_r) dA d\theta_r = F_0 \sum_i \sum_j f(\theta_{ri}) P_j(\theta_{ri}) \Delta A_j(\theta_i) \Delta \theta_i, \qquad \text{(D24)}$$

and

$$F_0 \int_{-\pi/2}^{+\pi/2} \int_{0}^{A(P_m,\theta_r)} dA d\theta_r = F_0 \sum_i \sum_j f(\theta_{ri}) \Delta A_j(\theta_r), \qquad \text{(D25)}$$

where the $f(\theta_{ri})$ were calculated from the Kessler debris flux model for relative angles, θ_{ri}, of /15°, 30°, 45°, 60°, and 75° for $\theta_{ri} = 15_0$, $\Delta A_j = (A_j - A_{j-1})$ with $A_j = \pi U_j W_j$ where U_j and W_j are vertical and horizontal miss distances associated with the collision probabilities, $P_j(\theta_{ri})$, computed either by Monte Carlo or numerically. The $P_j(\theta_{ri})$ were calculated for penetration of a 60-meter sphere about the space vehicle, while "probability of collision" in figures 3 and 4 is based on a 2000 meter^2 cross section for a space vehicle from any direction, and an annual flux through sphere of 1.6 passages/km^2 cross section.

It is informative to examine fractional residual risk as a function of P_m . This is shown in figure D-III.

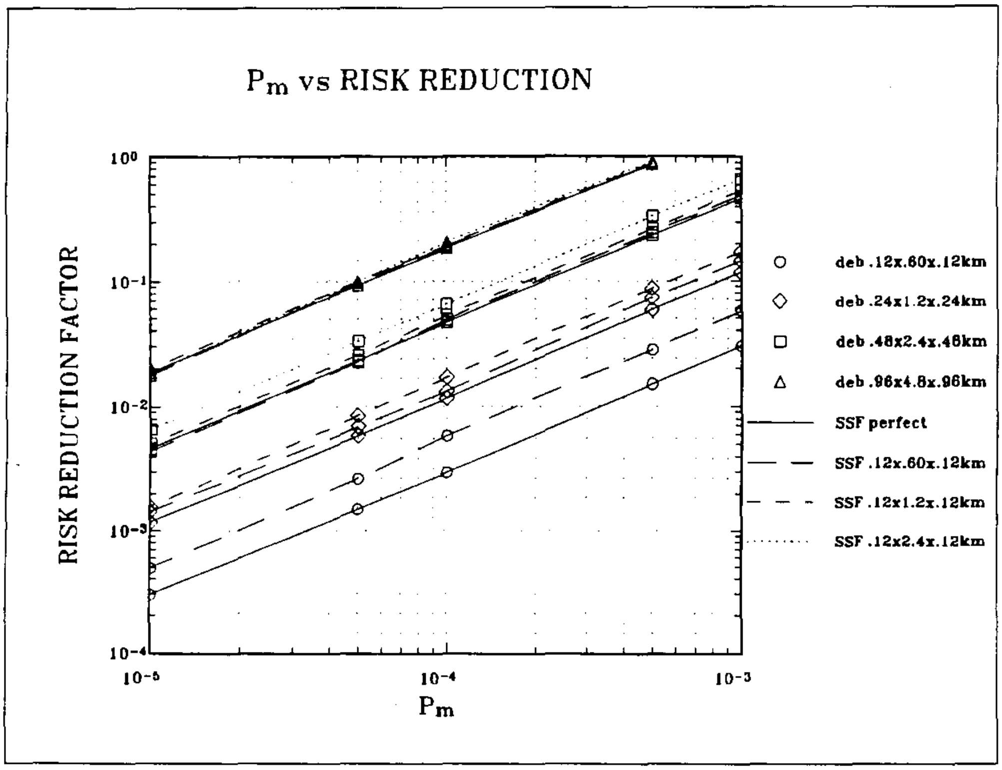

Figure D-III. Fractional residual risk or risk reduction factor as a function of P_m.

Appendix E
Maneuver Ellipsoids

For space vehicle and debris moving parallel to the earth, the surface defined by tangent debris trajectories having equal collision probabilities with the space vehicle target sphere approximately defines an ellipsoid. Specifically, given space vehicle/debris uncertainties and a collision probability for maneuver, P_m, the locus of cpa points about the vehicle with $P = P_m$ describes an ellipsoidal surface. Penetration into this ellipsoid by the class of debris under consideration will imply $P > P_m$ and that a debris avoidance maneuver should be performed.

When the probability of a collision, P, for a given set of uncertainty ellipsoids and given conjunction geometry is much smaller than the probability of a collision, P_0, for a predicted dead-on hit, the surfaces of constant P are approximately ellipsoidal. When $P \approx P_0$, the form of the surface defined by a constant P no longer resembles an ellipsoid even though the projection of constant P, from any direction θ_r , forms an ellipse. This section is included in the appendix because it is very important to see the spatial dimensions associated with the P_m . When $P_m \approx P_0$, a true ellipsoid is artificially generated for the figures in this section. In the opinion of the authors of this work, all ellipsoid axes shown are those which would be chosen, for these specific uncertainty parameterizations, should a maneuver criteria based on an ellipsoid be adopted.

The vertical axis is of the maneuver ellipsoid changes little with θ_r . This section of the appendix deals with obtaining maneuver ellipsoid horizontal axes from miss distance versus P_m data. Consider figure E-I below.

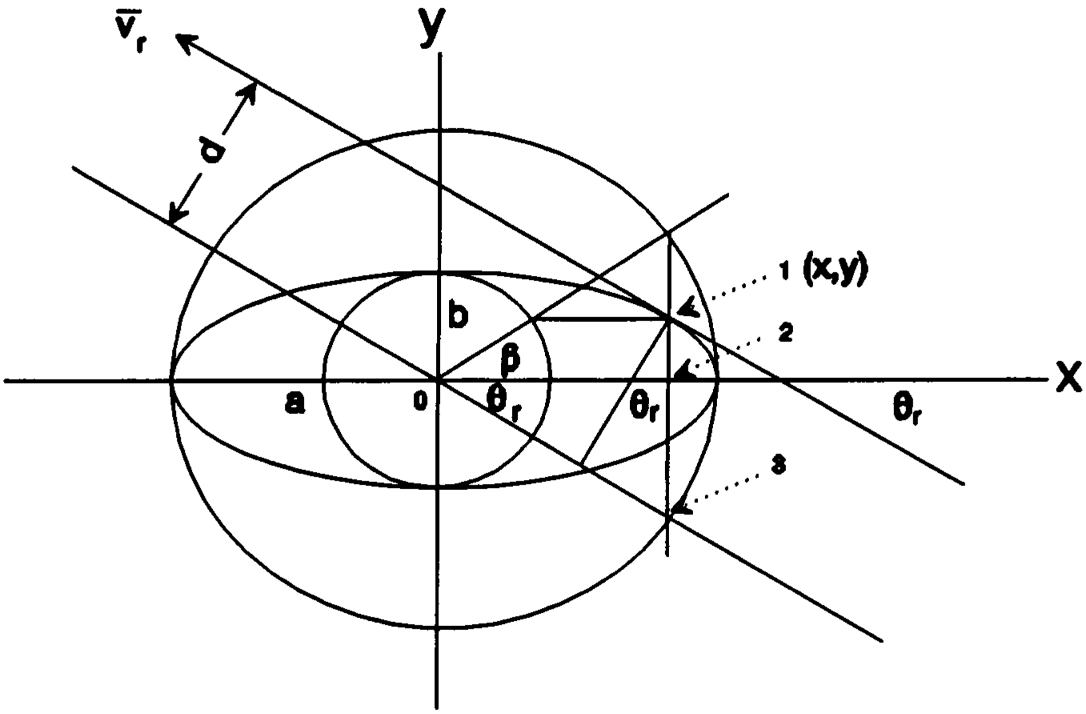

Figure E-I. Geometry in horizontal plane for debris coming toward the space vehicle at a relative angle θ_r with horizontal miss distance d.

Given the horizontal miss distances $\{d_i\}$ corresponding to a P_m, we wish to determine semimajor and semiminor axes in the horizontal plane. From the figure, the coordinates x,y of point 1 on the ellipse are given by

$$x = a \cos\beta \quad \text{and} \quad y = b \sin\beta \tag{E1}$$

where a and b are the radii of the inner and outer circles in figure E-I. The equation of the ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \; . \tag{E2}$$

The eccentricity, e, is

$$e = \sqrt{\frac{a^2 - b^2}{a^2}} \; . \tag{E3}$$

By defination

$$\tan\theta_r = \frac{dy}{dx} \quad . \tag{E4}$$

Combining equations E1, E2, E3, and E4 it is easily shown that

$$\tan\beta = \frac{\sqrt{1-e^2}}{\tan\theta_r} \quad . \tag{E5}$$

On figure E-I, points 0, 1, 2, and 3 are identified, defining lengths L_{02}, L_{12}, L_{23}, and L_{13}. The following relationships are seen:

$$L_{02} = x = a \cos\beta \quad \text{and} \quad L_{12} = y = b \cos\beta \ ,$$

$$\begin{gathered} \tan\theta_r = \frac{L_{23}}{L_{02}} \quad \Rightarrow \quad L_{23} = L_{02}\tan\theta_r \\ \text{and} \\ \cos\theta_r = \frac{d}{L_{13}} \quad \Rightarrow \quad L_{13} = \frac{d}{\cos\theta_r} \ . \end{gathered} \tag{E6}$$

Then

$$L_{13} = L_{12} + L_{23} = \frac{d}{\cos\theta_r} = y + a \cos\beta \tan\theta_r \quad . \tag{E7}$$

Data were taken at 5 angles θ_{ri} with 5 miss distances, d_i , corresponding to a given P_m. Multiplying equation E7 by $\cos\theta_{ri}$ and using relationships in equations E6,

$$\begin{gathered} d_i = y \cos\theta_{ri} + a \cos\beta_i \sin\theta_{ri} = \\ b \sin\beta_i \cos\theta_{ri} + a \cos\beta_i \sin\theta_{ri} = f(a,b) \quad . \end{gathered} \tag{E8}$$

The problem reduces to finding the a and b which minimize

$$\sum_i (d_i - f_i(a,b))^2 \ , \tag{E9}$$

which is the classic least squares problem.

The UVW axes are defined as having u up, v along the velocity vector, and w in the cross-track direction. Given the UVW axes, c, a, b of the ellipsoid corresponding to a constant hit probability, its equation is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \ . \tag{E10}$$

Given a relative position vector at cpa, $\vec{\rho}$, then

$$\vec{\rho} = A\hat{i} + B\hat{j} + C\hat{k} \ . \tag{E11}$$

If

$$\frac{A^2}{a^2} + \frac{B^2}{b^2} + \frac{C^2}{c^2} < 1 \ , \tag{E12}$$

the cpa is within the maneuver ellipsoid.

Figures E-II, E-III, and E-IV show the axes for the maneuver ellipsoids defined by the parameterization in this work. The lines on the graphs do not represent the actual paths between points.

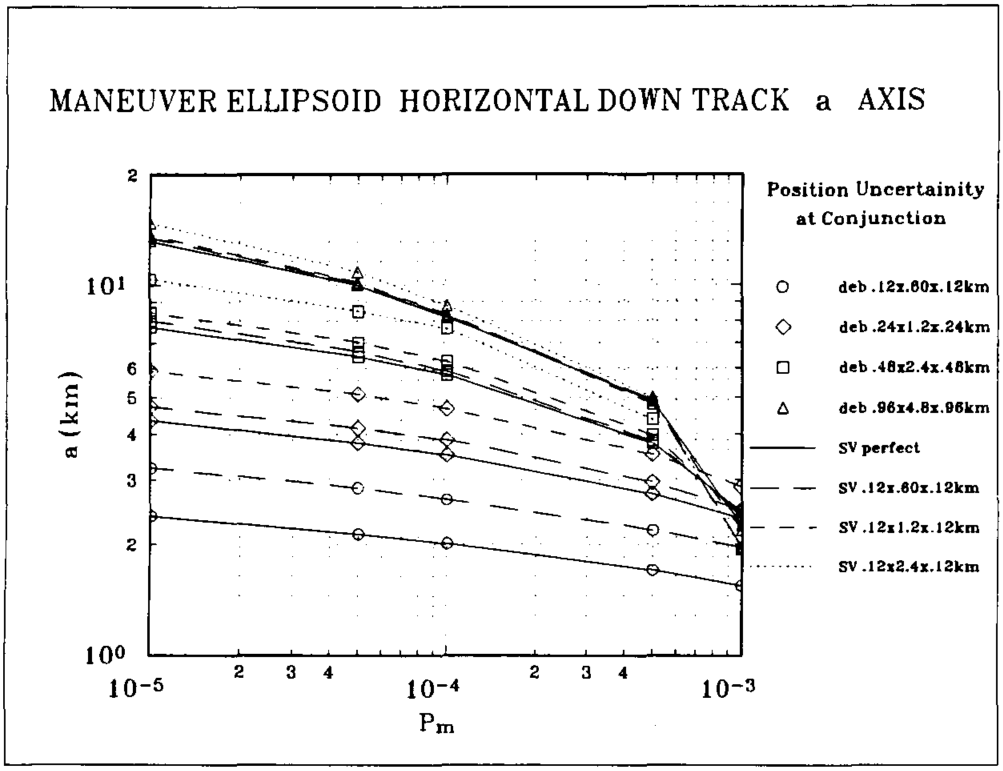

Figure E-II. Maneuver ellipsoid horizontal down-track "a" axis.

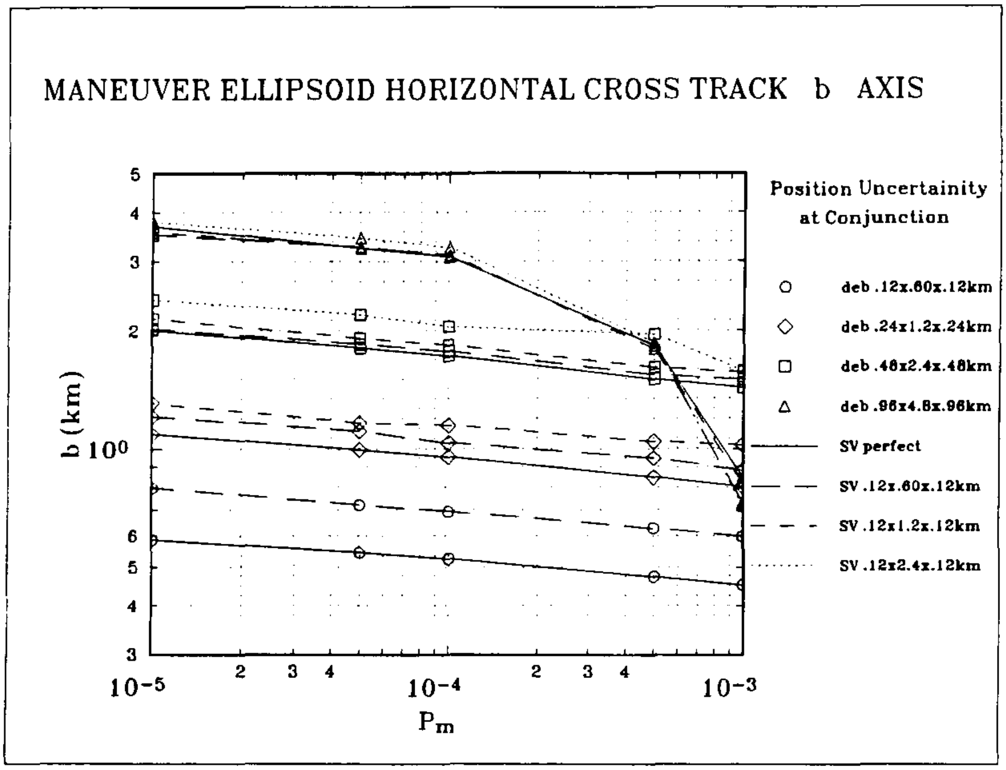

Figure E-III. Maneuver ellipsoid radial "b" axis.

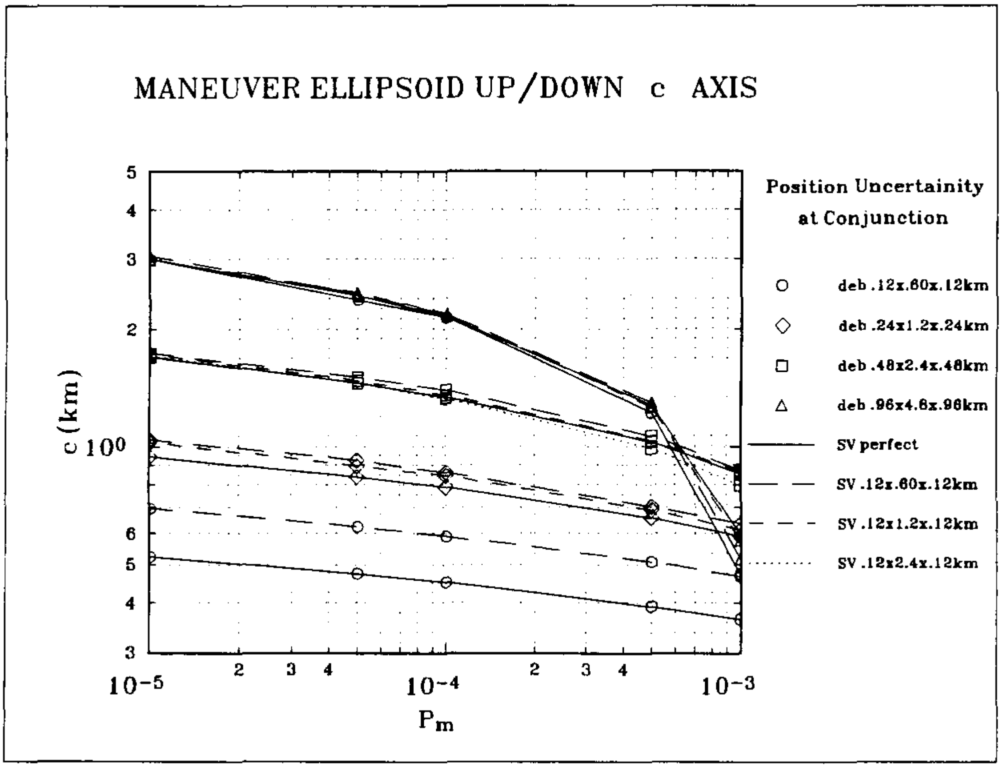

Figure E-IV. Maneuver ellipsoid up/down "c" axis.

Appendix F
Computation of Number of 30-Day Microgravity Periods

Current thinking is to define a space station program requirement for debris avoidance in terms of a maximum acceptable annual risk and a maximum number of expected debris avoidance maneuvers. These quantities, if accepted, will initiate additional system requirements and could compromise existing microgravity requirements. This appendix defines a Monte Carlo concept and defines equations used to produce data needed to evaluate the proposed requirements' effect on microgravity.

Monte Carlo Concept

The problem is to statistically determine the probability of having N consecutive 30-day periods of quiescence (no maneuvers) during a year. There are 4 planned reboost maneuvers per year at 90-day intervals, set, for example, on days 91, 182, 273 and 364. Debris avoidance maneuvers are not planned and occur randomly throughout the year. If M debris avoidance maneuvers are expected each year then the probability of a random maneuver occurring on a given day is M/365. Each of 364 days is simulated by computing a random number between 0 and 1; if the random number is less than 1/365, then a collision avoidance conjunction is defined to occur. The number of 30-day quiescent periods is determined by interrogating the maneuver flag for each day of the year to see whether or not a scheduled or unscheduled maneuver occurred. The procedure is repeated a million times to get a good statistical sample. The results of the procedure for six anticipated conjunctions warranting a maneuver in a year are presented in figure F-I.

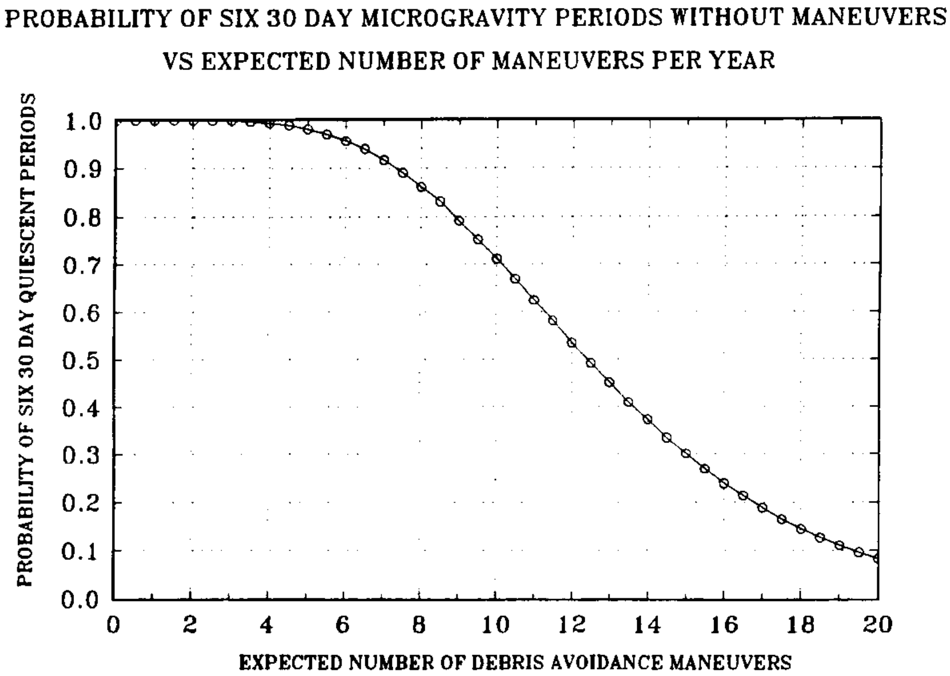

Figure F-I. Probability of six 30-day maneuver free periods as a function of anticipated maneuver rate.

space debris
Collision avoidance
independent variables
analyzing

space shuttles
Spacecraft

Distribution

NASA JSC
AT/Loftus, J. P.
DA/Stone, B. R.
DA4/Bourgeois, L. S.
DA14/Kennedy, M. G.
Lewis, C. R.
DE/Webb, D. J.
DE3/Stein, J. M.
DE33/Hemmer, P. M.
Kirk, R. L.
Shreve, J. V.
DE4/Ferring, M. J.
DH12/Gerstenmaier, W.
DI/Guillory, T. A.
Thompson, E. W.
DI21/Algate, A. F.
DI26/Jantzi, P. D.
Nader, B. A.
DI31/Snyder, R. D.
DI32/Shinkle, G. L.
Zimmerman, K. D.
DJ/Soileau, K. M.
DM/Deiterich, C. F.
Harpold, J. C.
DM2/Davis, L. D.
DM22/Davis, S. P.
DM24/Lowes, F. B.
Stein, R. T.
DM3/Smith, E. E.
DM32/Siders, J. H.
DM33/Gavin, R. T.
DM34/Evans, M. E.
Lunde, A. N.
Lostak, D. R.
Parker, C. B.
Pido, K. I.
Presley, W. S.
Sanchez, R.
Williamson, M. L.
DM4/Collins, M. F.
DM42/Pearson, D. J.
Stich, S. J.
Haynes, M.
Adamo, D.
DM43/Epp, C. D.
DM46/Oliver, G. T.
EA/Blevins, D.
EE6/Arndt, R.
Dekome, K.
Saunders, P. E.
EG/Cox, K. J.
Kramer, P. C.
EG4/Hite, G. C.
McHenry, L.
Yeo, J. E.
EG42/Blucker, T. J.
EG43/Estes, H. S.
Jackson, W. L.
Le, T. Q.
ET/Graves, C. A.
ET13/Neider, R. L.
ET3/Hansen, L. E.
McNeely, J. T.
KA/Bobola, R. E.
KC/Haines, D. C.
SA/White, T. T.
SN3/Potter, A.
SN31/Christiansen, E. L.
Kessler, D. J.
Vilas, F.
TA/Lambert, Jr., C. H.
Smothermon, J. L.
VA/Germany, D. M.

NASA Reston
DSS-2/Cramer, B.
DS/Moorehead, R.
DSS/Thorson, R. A.

STSOC
R14A/Contella, M. C.
Osburn, R. K.
R16C/Anderson, S. A.
Becker, R. W.
Burton, W. C.
Conway, H. L.
Elk, J. R.
Oberg, J. E.
Parker, C. R.

OSC
BARR08/Carley, K. W.
Endalwaght, C. R.
Ewan, P. V.
Foster, J. L. (55)
Frament, M. A.
Powell, M. A.
DM34/Zoerner, G. D.
DE33/Cristol, S. C.
DI32/Alexander, J. D.
R16M/Schultz, E. D.
R55A/Seay, W. A.

SEIC Reston
364-313/Bloom, B.
Lewis, T.
Malloy, W.
Rogan, J.E.
West, S.

SEIC Houston
Young, K.
Thorn, V.

AASC
Bond, V.
Semar, C.
Tabor, J.
Vedder, J.

WP-2 MDSSC Houston
Murdock, K.
Stegall, H.

LOCKHEED
C23/Anz-Medor, P. D.
Reynolds, R. C.
Talent, D.

TRW
Baxter, J.
Henderson, D.
Lee, R.